Answer Key & Teacher Notes

Session 1 — Surface Area of Pyramids · Session 2 — Lateral Area of Pyramids

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Surface Area of Pyramids

TALK PROMPTS

- Why do you use the slant height instead of the pyramid's straight-up height?
- Which measure belongs to the square, and which one belongs to the triangles?

PARTNER TASK

Pyramid Tent. A tent has a square base 6 ft on each side and a slant height of 5 ft. One partner finds the base (36 ft²), the other one triangle (15 ft²). Together decide how many triangles to add (36 + 60 = 96 ft²), then decide what changes if the floor is open ground (60 ft²).

WATCH FOR

- Adding the base and only one triangle.
- Using the pyramid's straight-up height instead of the slant height.
- Forgetting the ½ in each triangle's area.

SESSION 2 · TEACHER NOTES

Lateral Area of Pyramids

TALK PROMPTS

- When should you include the base in a total, and when should you leave it out?
- What words in a problem tell you to leave the base out?

PARTNER TASK

Two Pyramids. Both have a base 9 in on each side and a slant height of 12 in. Partner A paints only the slanted faces (216 in²); Partner B paints every face (297 in²). Together explain why the answers differ by exactly 81 in².

WATCH FOR

- Adding the base to a lateral area when the base is not covered.
- Using 4 slanted faces for a pyramid with a triangle base (it has 3).
- Forgetting the ½ in each triangle's area.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.G.A.4	1	1	1	1	2	2	1	3	2	1	1
S1 · Form B	6.G.A.4	1	1	1	1	2	2	1	3	2	1	1
S2 · Form A	6.G.A.4	1	2	1	1	1	2	2	3	2	1	1
S2 · Form B	6.G.A.4	1	2	1	1	1	2	2	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

slant height: The height of a side triangle, measured **along** its face.

lateral face: A side **triangle** of a pyramid.

apex: The **top** point where the side faces meet.

- SA of a pyramid = base area + area of all the **triangular** faces.
- A square pyramid has **4** matching triangles.
- Use the **slant** height, not the straight-up height, for the triangles.

Example: Base: 10 × 10 = **100** cm². One triangle: ½ × 10 × 7 = **35** cm². Four triangles: 4 × 35 = **140** cm². Answer: SA = 100 + 140 = **240** cm²

YOU TRY

1. 16 + 4 × (½ × 4 × 6) = 16 + 48 = 64 m²

2. a. ½ × 6 × 5 = 15 in² b. 36 in²

EXIT TICKET

1. a. 81 cm² b. ½ × 9 × 4 = 18 cm²

2. 36 + 4 × (½ × 6 × 7) = 36 + 84 = 120 in²

3. **A** (the slant height)

PRACTICE FORM A

1. a. 64 + 4 × 40 = 224 m² b. 25 + 4 × 20 = 105 ft² c. 4 + 4 × 3 = 16 cm² d. 49 + 4 × 35 = 189 in²

2. a. 16 × 16 = 256 cm² b. ½ × 16 × 15 = 120 cm² c. 256 + 4 × 120 = 736 cm²

3. 64 + 4 × (½ × 8 × 11) = 64 + 176 = 240 in²

4. 144 + 4 × (½ × 12 × 9) = 144 + 216 = 360 cm²

5. Base 20 × 20 = 400 ft²; triangles 4 × (½ × 20 × 26) = 1,040 ft²; SA = 1,440 ft²

6. No. 117 cm² is one triangle. SA = 324 + 4 × 117 = 792 cm².

7. 256 + 4 × (½ × 16 × 10) = 256 + 320 = 576 in²

8. a. base 196 in²; one triangle ½ × 14 × 15 = 105 in² b. 196 + 4 × 105 = 616 in² c. 420 × 0.04 = $16.80

9. Ethan added only one triangle. A square pyramid has 4 triangles: 4 × 187 = 748. SA = 484 + 748 = 1,232 in².

10. **B** (4)

11. **D** (105 cm²)

Self-check word **PYRAMIDS**

PRACTICE FORM B

1. a. 36 + 4 × 24 = 132 m² b. 16 + 4 × 14 = 72 ft² c. 9 + 4 × 7 1/2 = 39 cm² d. 81 + 4 × 45 = 261 in²

2. a. 14 × 14 = 196 cm² b. ½ × 14 × 12 = 84 cm² c. 196 + 4 × 84 = 532 cm²

3. 100 + 4 × (½ × 10 × 13) = 100 + 260 = 360 in²

4. 64 + 4 × (½ × 8 × 9) = 64 + 144 = 208 cm²

5. Base 24 × 24 = 576 ft²; triangles 4 × (½ × 24 × 20) = 960 ft²; SA = 1,536 ft²

6. No. 88 cm² is one triangle. SA = 256 + 4 × 88 = 608 cm².

7. 144 + 4 × (½ × 12 × 14) = 144 + 336 = 480 in²

8. a. base 100 in²; one triangle ½ × 10 × 12 = 60 in² b. 100 + 4 × 60 = 340 in² c. 240 × 0.05 = $12

9. Mia added only one triangle. A square pyramid has 4 triangles: 4 × 150 = 600. SA = 400 + 600 = 1,000 in².

10. **B** (5)

11. **C** (96 cm²)

Self-check word **MONUMENT**

GUIDED NOTES

lateral area: The area of the slanted faces **only**.

surface area: The area of **every** face, base included.

lateral face: A **slanted** side triangle of a pyramid.

- Lateral area = the area of all the **slanted** faces.
- Surface area = lateral area + **base** area.
- Read the problem to decide whether the base is **covered**.

Example: Cover the 4 slanted triangles. Leave out the **base**. One triangle: ½ × 4 × 6 = **12** m². Four triangles: 4 × 12 = **48** m². Answer: lateral area = **48** m²

YOU TRY

1. 4 × (½ × 6 × 7) = 4 × 21 = 84 ft²

2. a. 90 − 25 = 65 m² b. 3 × 12 = 36 in²

EXIT TICKET

1. a. 60 cm² b. 100 − 36 = 64 in²

2. 4 × (½ × 10 × 6) = 4 × 30 = 120 m²

3. **A** (lateral area)

PRACTICE FORM A

1. a. 4 × 25 = 100 cm² b. 150 − 36 = 114 in² c. 4 × (½ × 7 × 9) = 126 m² d. 4 × 55 = 220 in²

2. a. 3 b. ½ × 24 × 14 = 168 in² c. 3 × 168 = 504 in²

3. a. ½ × 10 × 9 = 45 ft² b. 4 × 45 = 180 ft² c. 180 + 100 = 280 ft²

4. 4 × (½ × 14 × 8) = 4 × 56 = 224 ft²

5. 4 × (½ × 22 × 17) = 4 × 187 = 748 m²

6. 676 + 4 × (½ × 26 × 25) = 676 + 1,300 = 1,976 in²

7. No. Kai forgot the ½. One face is ½ × 30 × 21 = 315 cm², so the lateral area is 4 × 315 = 1,260 cm².

8. a. 4 × (½ × 16 × 10) = 320 ft² b. 320 × 3 = $960 c. 320 + 256 = 576 ft²; the floor is not covered, so the lateral area is the canvas needed.

9. Rosa added the base (196 cm²), but the base is not painted. The painted area is the lateral area only: 4 × (½ × 14 × 8) = 224 cm².

10. **A** (base)

11. **A** (3)

Self-check word **SHINGLES**

PRACTICE FORM B

1. a. 4 × 30 = 120 cm² b. 180 − 64 = 116 in² c. 4 × (½ × 5 × 8) = 80 m² d. 4 × 54 = 216 in²

2. a. 3 b. ½ × 18 × 10 = 90 in² c. 3 × 90 = 270 in²

3. a. ½ × 12 × 10 = 60 ft² b. 4 × 60 = 240 ft² c. 240 + 144 = 384 ft²

4. 4 × (½ × 16 × 9) = 4 × 72 = 288 ft²

5. 4 × (½ × 18 × 15) = 4 × 135 = 540 m²

6. 400 + 4 × (½ × 20 × 24) = 400 + 960 = 1,360 in²

7. No. Luis forgot the ½. One face is ½ × 24 × 15 = 180 cm², so the lateral area is 4 × 180 = 720 cm².

8. a. 4 × (½ × 12 × 8) = 192 ft² b. 192 × 4 = $768 c. 192 + 144 = 336 ft²; the floor is not covered, so the lateral area is the canvas needed.

9. Grace added the base (144 cm²), but the base is not painted. The painted area is the lateral area only: 4 × (½ × 12 × 7) = 168 cm².

10. **A** (base area)

11. **B** (4)

Self-check word **COVERAGE**

SHEET 1 · SURFACE AREA OF PYRAMIDS

1. $36 + 4 \times (\frac{1}{2} \times 6 \times 9) = 36 + 108 = 144$ m²
2. a. 144 cm² b. $\frac{1}{2} \times 12 \times 10 = 60$ cm² SA = 144 + 240 = 384 cm²
3. a. $16 + 4 \times (\frac{1}{2} \times 4 \times 5) = 56$ in² b. $100 + 4 \times (\frac{1}{2} \times 10 \times 12) = 340$ cm²
4. $81 + 4 \times (\frac{1}{2} \times 9 \times 11) = 81 + 198 = 279$ in²
5. He used the pyramid's height (4 ft) instead of the slant height (5 ft). SA = $36 + 4 \times (\frac{1}{2} \times 6 \times 5) = 36 + 60 = 96$ ft².

SHEET 2 · LATERAL AREA OF PYRAMIDS

1. $4 \times (\frac{1}{2} \times 8 \times 6) = 96$ ft²
2. a. 136 in² b. 72 cm² c. $4 \times (\frac{1}{2} \times 6 \times 10) = 120$ m² d. $4 \times (\frac{1}{2} \times 3 \times 4) = 24$ ft²
3. Lateral area $4 \times (\frac{1}{2} \times 14 \times 10) = 280$ ft²; 280 ÷ 35 = 8 bundles
4. One face: 160 ÷ 4 = 40 in²; $\frac{1}{2} \times 8 \times$ slant height = 40, so slant height = 10 in
5. No. 85 ft² = 25 + 60 includes the floor. With no floor, only the lateral area is needed: $4 \times (\frac{1}{2} \times 5 \times 6) = 60$ ft².
6. **B** (56 cm²)
7. **C** (36 m²)

October 2026 conceptual additions — teacher key

Session 1 · Surface Area of Pyramids

Base 36 cm²; each side triangle 15 cm²; four side triangles 60; total 96 cm². Slant height is the perpendicular height within a lateral triangle, not the solid's vertical height.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: Base 36 cm²; each side triangle 15 cm²; four side triangles 60; total 96 cm². Slant height is the perpendicular height within a lateral triangle, not the solid's vertical height.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Lateral Area of Pyramids

Lateral area 4 × 1/2 × 8 × 5 = 80 m². Complete surface adds base 64, giving 144 m². The roof-covering situation excludes the base.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: Lateral area 4 × 1/2 × 8 × 5 = 80 m². Complete surface adds base 64, giving 144 m². The roof-covering situation excludes the base.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.